

Asthma

Asthma is a chronic inflammatory disease of the airways in which the bronchial tubes become swollen and narrowed, making it difficult for air to move in and out of the lungs. The inflammation also causes increased mucus production and makes the airways extremely sensitive to triggers such as dust, pollen, cold air, smoke, exercise, or respiratory infections. During an asthma attack, the muscles surrounding the airways tighten, further reducing airflow and causing wheezing and shortness of breath.

Patients commonly complain of recurrent episodes of wheezing, chest tightness, persistent coughing, particularly during the night or early morning, and difficulty breathing. Symptoms may disappear completely between attacks or remain persistent depending on disease severity.

A chest X-ray is usually normal in patients with asthma. During severe attacks, the lungs may appear hyperinflated because excess air becomes trapped inside them. Since asthma often produces a normal chest X-ray, diagnosis mainly relies on pulmonary function tests, spirometry, clinical history, and response to bronchodilator medications.

Treatment focuses on reducing airway inflammation and preventing attacks. Inhaled corticosteroids remain the most effective long-term therapy, while short-acting bronchodilators provide rapid relief during acute attacks. Patients are advised to avoid environmental triggers, stop smoking, maintain a healthy lifestyle, and seek immediate medical attention if breathing difficulty does not improve after using a rescue inhaler.

Chronic Obstructive Pulmonary Disease (COPD)

Chronic Obstructive Pulmonary Disease is a progressive lung disease characterized by persistent airflow limitation that is usually irreversible. The disease commonly develops after many years of cigarette smoking, although prolonged exposure to biomass fuel smoke, industrial dust, and air pollution may also contribute. COPD consists primarily of chronic bronchitis and emphysema, both of which gradually reduce lung function over time.

Patients typically experience chronic cough, excessive sputum production, increasing shortness of breath during physical activity, wheezing, fatigue, and frequent respiratory infections. As the disease advances, oxygen levels decrease, and patients may develop respiratory failure or heart complications.

Chest X-rays often demonstrate hyperinflated lungs, flattened diaphragms, increased retrosternal air space, enlarged lungs, and sometimes bullae resulting from emphysema. Pulmonary function tests confirm persistent airflow obstruction.

Management includes smoking cessation, bronchodilator inhalers, inhaled corticosteroids when indicated, pulmonary rehabilitation, supplemental oxygen for advanced disease, vaccination against influenza and pneumococcal infection, and regular medical follow-up.

Lung Cancer

Lung cancer is one of the leading causes of cancer-related deaths worldwide. It develops when abnormal cells within the lungs grow uncontrollably and form malignant tumors capable of invading surrounding tissues or spreading to distant organs. Cigarette smoking remains the strongest risk factor, although exposure to asbestos, radon gas, industrial chemicals, and air pollution also increases risk.

Early-stage lung cancer often causes no symptoms. As the disease progresses, patients may develop persistent cough, coughing up blood, chest pain, unexplained weight loss, hoarseness, fatigue, recurrent pneumonia, and progressive shortness of breath.

Chest X-rays may reveal solitary pulmonary nodules, irregular lung masses, hilar enlargement, collapse of lung tissue, pleural effusion, or enlarged mediastinal lymph nodes. However, CT scanning provides greater sensitivity for detecting small tumors.

Diagnosis requires imaging studies followed by biopsy to confirm malignancy.

Treatment depends on cancer type and stage and may involve surgery, chemotherapy, radiation therapy, targeted therapy, immunotherapy, or combinations of these approaches.

Pleural Effusion

Pleural effusion refers to the abnormal accumulation of fluid within the pleural cavity surrounding the lungs. It may result from heart failure, pneumonia, tuberculosis, malignancy, kidney disease, liver disease, or inflammatory disorders.

Patients commonly complain of progressive shortness of breath, chest discomfort, dry cough, and difficulty taking deep breaths. Large pleural effusions significantly reduce lung expansion and impair oxygen exchange.

Chest X-ray typically demonstrates blunting of the costophrenic angle, homogeneous lower lung opacity, and fluid layering within the pleural cavity. Ultrasound and CT scanning help determine fluid volume and guide drainage procedures.

Treatment depends upon the underlying cause and may involve antibiotics, diuretics, drainage using thoracentesis, or surgical intervention.

Pneumothorax

Pneumothorax occurs when air enters the pleural cavity, causing partial or complete collapse of the affected lung. It may develop spontaneously, following chest trauma, mechanical ventilation, or rupture of lung blebs.

Patients experience sudden sharp chest pain, severe shortness of breath, rapid breathing, and reduced breath sounds on the affected side.

Chest X-ray demonstrates a visible pleural line with absence of lung markings beyond it and collapse of lung tissue. Large pneumothoraces require urgent drainage using a chest tube.

Pulmonary Fibrosis

Pulmonary fibrosis is a chronic lung disease characterized by progressive scarring of lung tissue. The scar tissue gradually replaces healthy lung tissue, reducing lung elasticity and impairing oxygen transfer into the bloodstream.

Patients usually develop persistent dry cough, increasing breathlessness during physical activity, fatigue, unexplained weight loss, and finger clubbing.

Chest X-ray may demonstrate diffuse reticular opacities, reduced lung volume, and coarse interstitial markings. High-resolution CT scanning is considerably more sensitive and often demonstrates honeycombing and fibrosis.

Although fibrosis cannot usually be reversed, antifibrotic medications, oxygen therapy, pulmonary rehabilitation, and lung transplantation may improve survival and quality of life.

Bronchitis

Bronchitis is inflammation of the bronchial tubes that carry air to and from the lungs. Acute bronchitis is commonly caused by viral infections, whereas chronic bronchitis usually develops after prolonged cigarette smoking.

Patients present with persistent cough, sputum production, wheezing, chest discomfort, fatigue, and mild fever. Acute bronchitis often resolves within a few weeks, whereas chronic bronchitis persists for months each year.

Chest X-rays are frequently normal or show mild bronchial wall thickening. Treatment includes hydration, rest, bronchodilators when necessary, smoking cessation, and antibiotics only when bacterial infection is strongly suspected.

Pulmonary Edema

Pulmonary edema occurs when fluid accumulates within the lung tissue and alveoli, most commonly because of heart failure. The excess fluid interferes with oxygen exchange and can rapidly become life-threatening.

Patients develop sudden shortness of breath, difficulty lying flat, coughing up frothy sputum, rapid breathing, anxiety, and low oxygen saturation.

Chest X-rays typically demonstrate bilateral diffuse opacities, enlarged cardiac silhouette, Kerley B lines, and pleural effusions. Treatment focuses on oxygen therapy, diuretics, management of heart failure, and emergency medical care.

Heart Enlargement (Cardiomegaly)

Cardiomegaly refers to enlargement of the heart, often resulting from hypertension, heart valve disease, cardiomyopathy, or chronic heart failure. Patients may experience fatigue, shortness of breath, swelling of the legs, and reduced exercise tolerance.

Chest X-ray reveals an enlarged cardiac silhouette with a cardiothoracic ratio greater than 50%. Echocardiography is generally required to determine the underlying

cardiac abnormality.

Treatment depends upon the underlying heart disease and may include blood pressure control, medications for heart failure, valve surgery, or lifestyle modifications.

Respiratory Syncytial Virus (RSV)

Respiratory Syncytial Virus is a highly contagious viral infection that primarily affects infants, older adults, and immunocompromised individuals. It causes inflammation of the small airways and may progress to bronchiolitis or pneumonia.

Symptoms include cough, wheezing, fever, nasal congestion, rapid breathing, and reduced feeding in infants. Chest X-rays may demonstrate hyperinflation, peribronchial thickening, and patchy infiltrates. Treatment is mainly supportive, with oxygen therapy reserved for severe disease